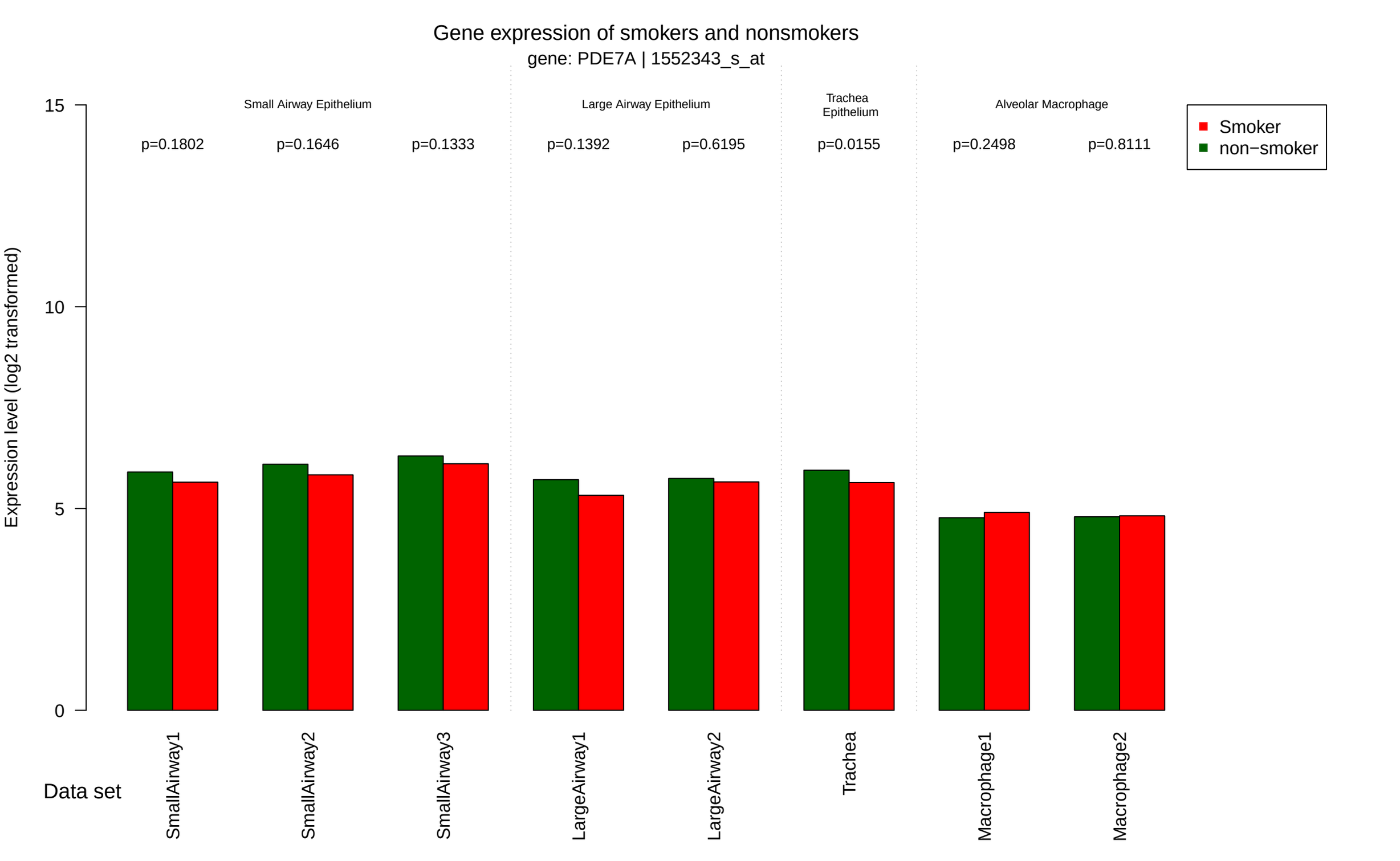
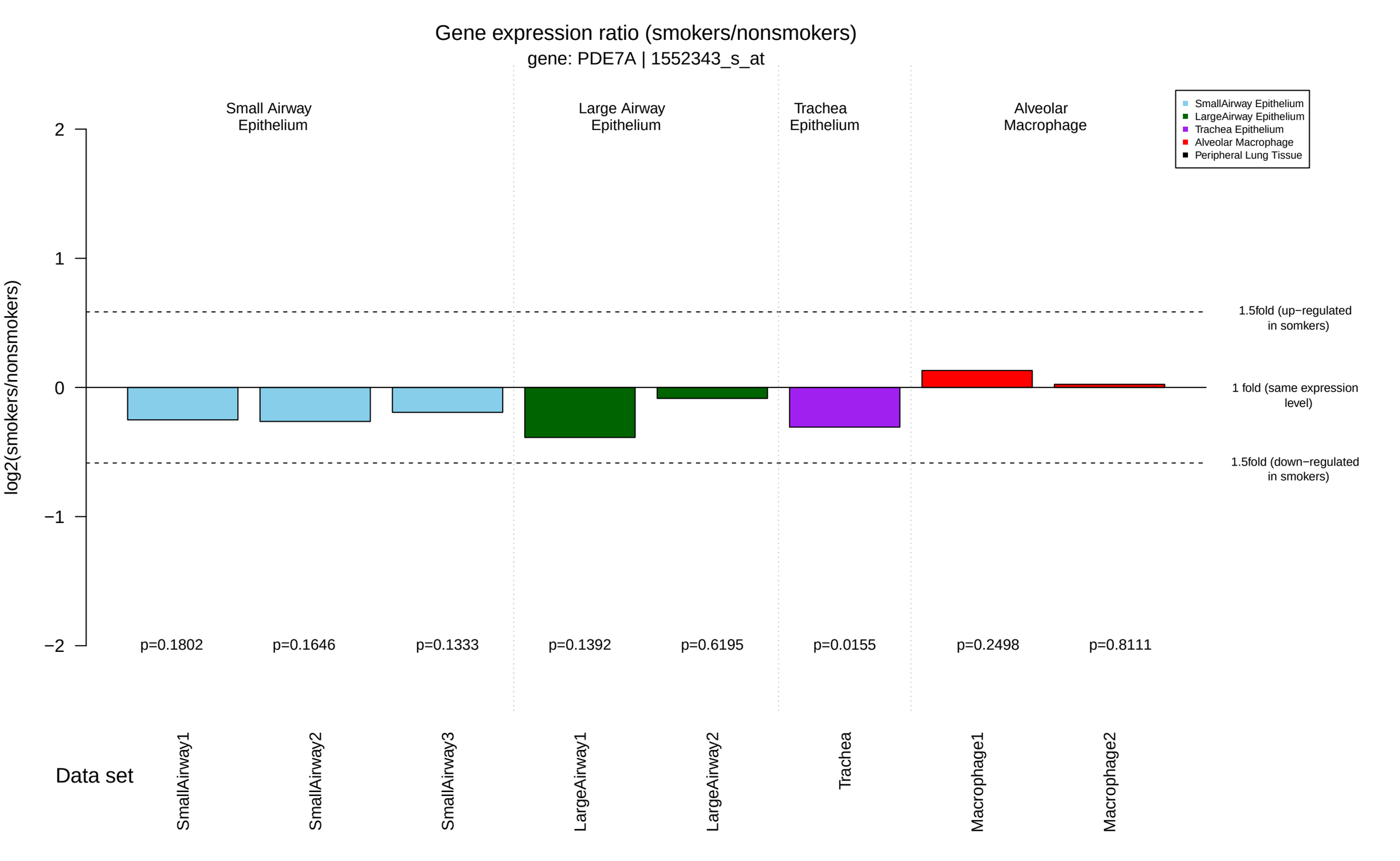
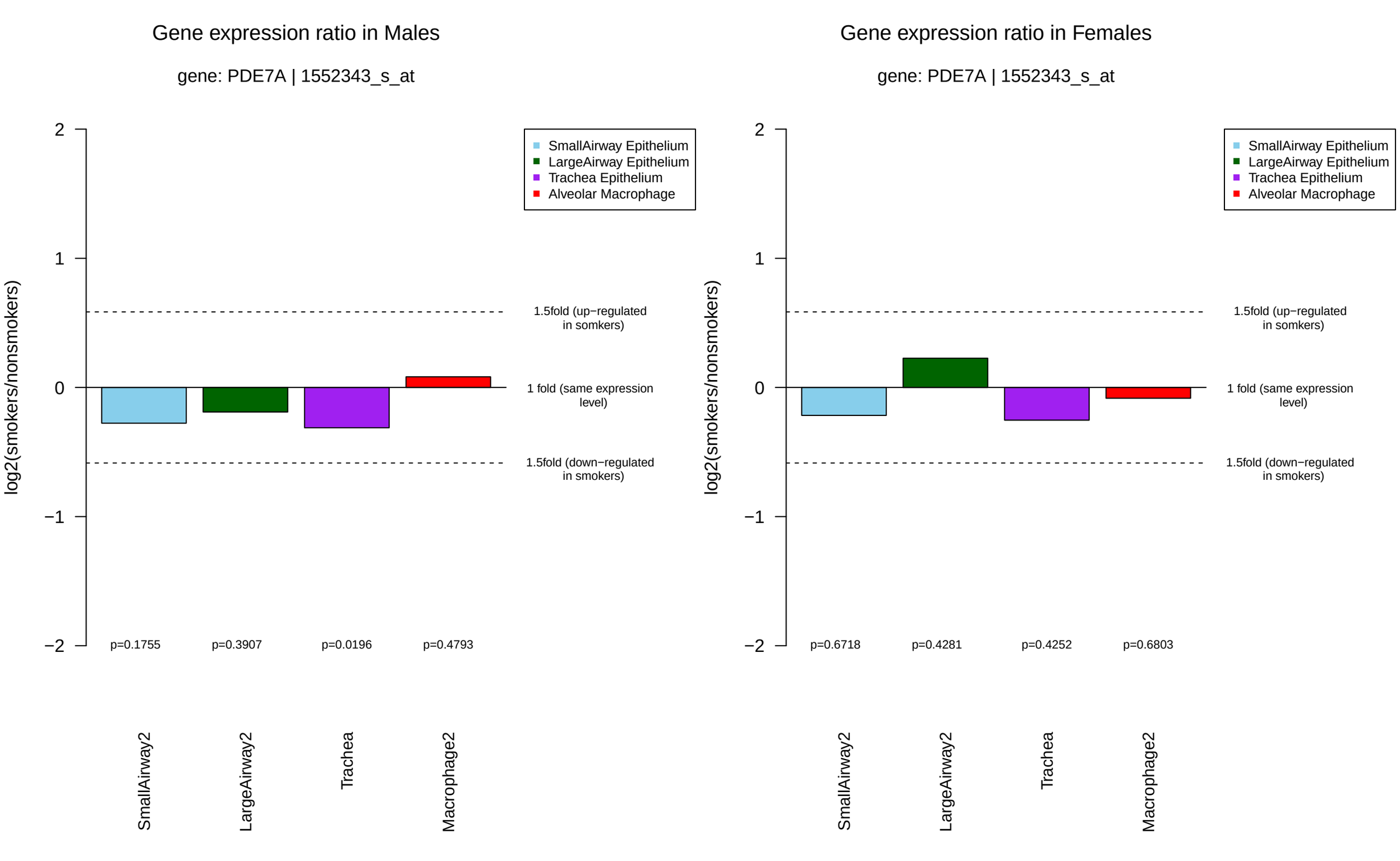
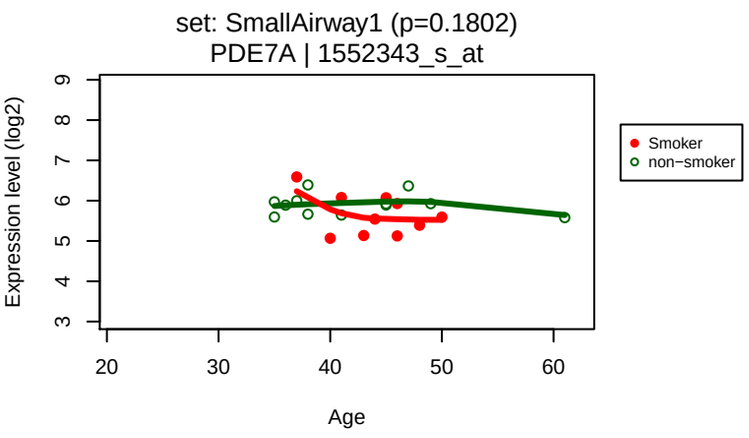
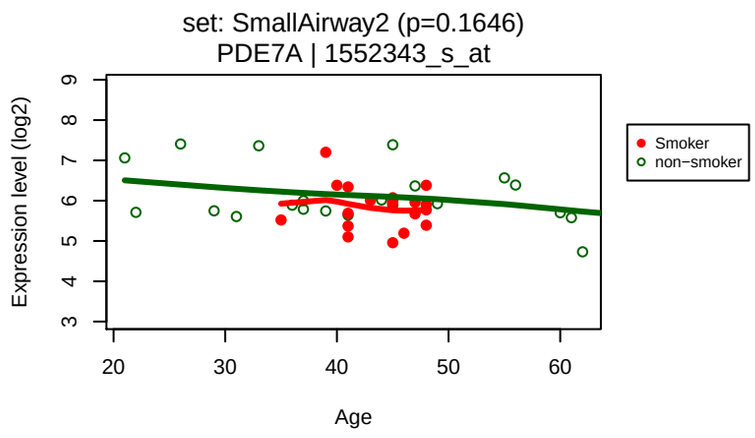
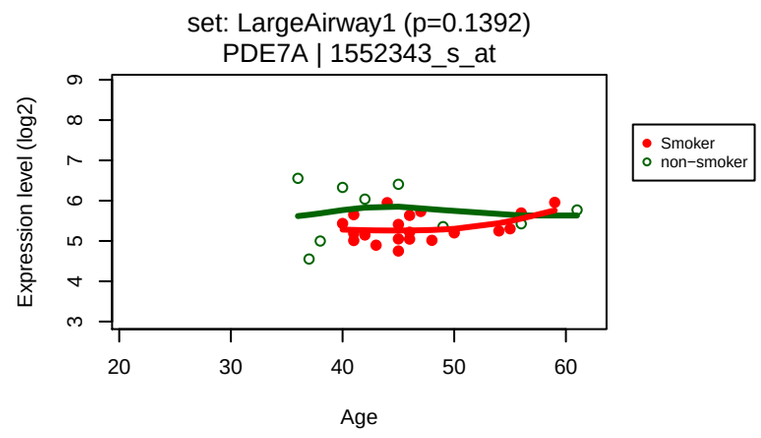

Gene expression of smokers and nonsmokers
gene: PDE7A | 1552343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1802
p=0.1646
p=0.1333
p=0.1392
p=0.6195
p=0.0155
p=0.2498
p=0.8111
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

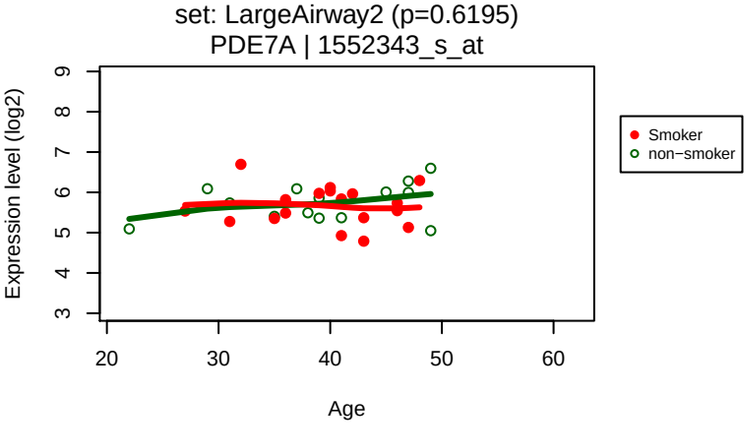

Gene expression ratio (smokers/nonsmokers)
gene: PDE7A | 1552343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1802
p=0.1646
p=0.1333
p=0.1392
p=0.6195
p=0.0155
p=0.2498
p=0.8111
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

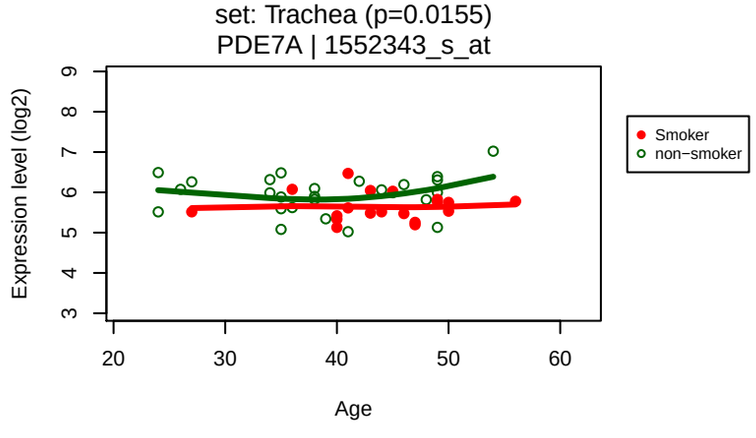

Gene expression ratio in Males
gene: PDE7A | 1552343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1755
p=0.3907
p=0.0196
p=0.4793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE7A | 1552343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6718
p=0.4281
p=0.4252
p=0.6803
SmallAirway2
LargeAirway2
Trachea
Macrophage2

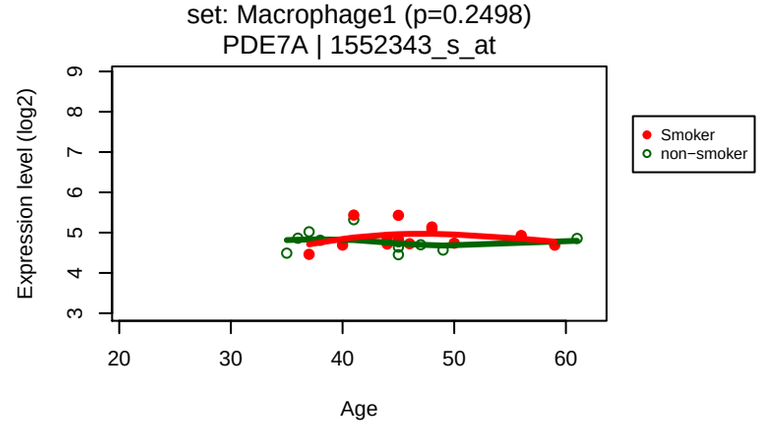

set: SmallAirway1 (p=0.1802)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker

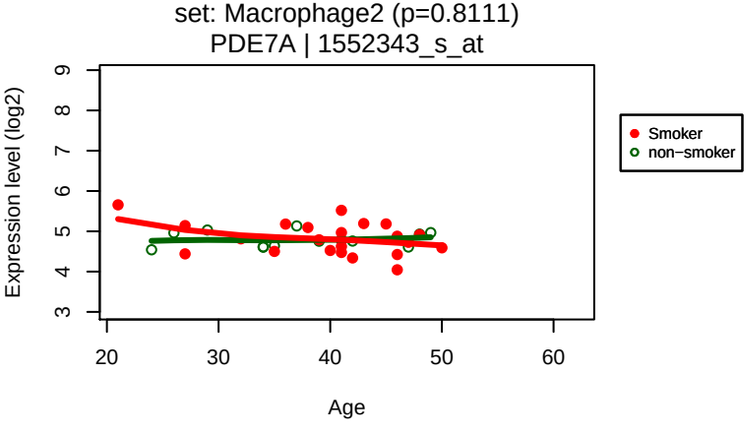

set: SmallAirway2 (p=0.1646)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker

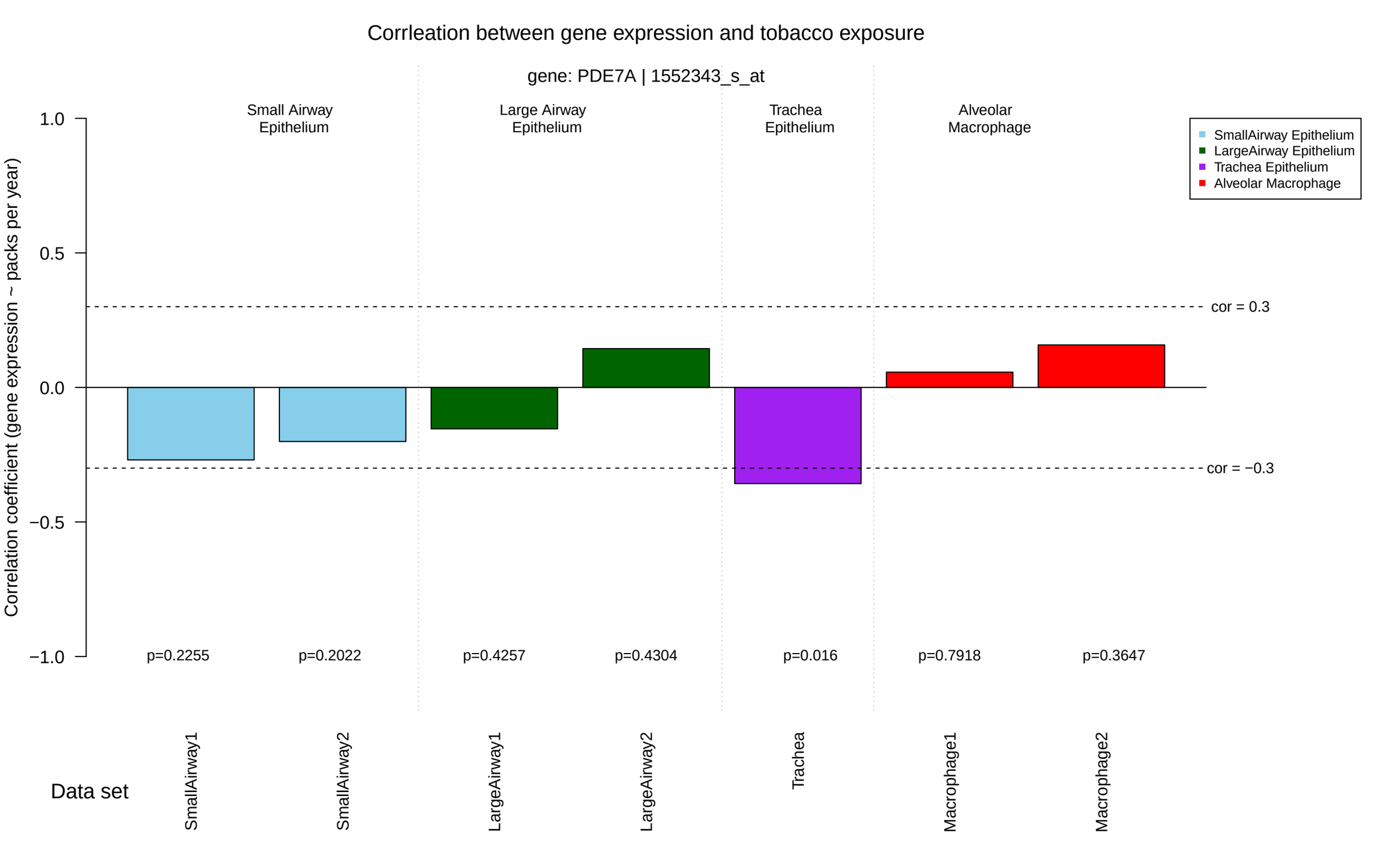

set: LargeAirway1 (p=0.1392)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6195)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0155)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2498)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8111)
PDE7A | 1552343_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE7A | 1552343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2255
p=0.2022
p=0.4257
p=0.4304
p=0.016
p=0.7918
p=0.3647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2